

UNIVERSIDADE FEDERAL DO AMAZONAS

PPGEE — Otimização Numérica

Relatório Final do Trabalho 4

Benchmarking de Solvers NLP — Ipopt, MadNLP, NLOpt, Optim, Uno

Versão corrigida — todos os solvers e problemas consolidados

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1. Introdução

Este relatório consolida o benchmark de Programação Não-Linear (NLP) em duas frentes: Parte 1 — 47 problemas PLATO ASU AMPL-NLP (convertidos de AMPL para JuMP); Parte 2 — 202 problemas da COCONUT Benchmark Library 1. Cada problema foi resolvido com cinco solvers. O melhor solver de cada problema é o que atinge a melhor função-objetivo (LOCALLY_SOLVED/ALMOST_LOCALLY_SOLVED), com desempate por tempo; o gap relativo é $|f_{\text{solver}} - f_{\text{best}}| / \max(1, |f_{\text{best}}|)$.

2. Metodologia

Resolução em Julia/JuMP. Parte 1: conjunto oficial de 47 problemas AMPL-NLP em JuMP; Parte 2: biblioteca COCONUT (GAMS) convertida para JuMP. Cada par (problema, solver) roda em processo Julia isolado (crash-safe), limite de 300 s. Resolvido = LOCALLY_SOLVED/OPTIMAL (ou ALMOST) com objetivo finito.

2.1 Configuração dos solvers

• **Ipopt**: max_iter 5000, tol 1e-6, com acceptable_tol 1e-2 (fallback). • **MadNLP**: max_iter 5000, tol 1e-6. • **NLopt**: LD_SLSQP. • **Uno**: preset 'filtersqp'. • **Optim***: sem wrapper JuMP para restrições gerais → executado por proxy com Ipopt (por isso coincide com o Ipopt).

2.2 Gap de otimalidade

Solvers locais, sem certificado global. Gap medido contra o melhor valor entre solvers (f_{best}): $\text{signed_gap} = f_{\text{solver}} - f_{\text{best}}$; $\text{relative_gap} = |f_{\text{solver}} - f_{\text{best}}| / \max(1, |f_{\text{best}}|)$. Reporta-se status, objetivo, f_{best} , gap absoluto/relativo e tempo por par.

3. Parte 1 — PLATO ASU AMPL-NLP (47 problemas)

3.1 Resumo por solver

Solver	Total	Resolv.	%	Gap $\leq 1e-4$	Gap $\leq 1e-6$	T.médio(s)	T.med.(s)
ipopt	47	30	63.8%	26	24	180.1	52.5
madnlp	47	27	57.4%	27	26	267.9	95.1
nlopt	47	0	0.0%	0	0	191.0	38.0
optim	47	30	63.8%	26	24	180.1	52.5
uno	47	0	0.0%	0	0	489.2	600.4

Distribuição do 'melhor solver': ipopt=14 | madnlp=17 | nlopt=0 | optim=0 | uno=0

3.2 Cobertura na melhor situação de solução

Situação (melhor caso)	Qtd	%
Resolvido (algum solver convergiu)	31	66.0%
Melhor incumbente (no time-limit)	4	8.5%
Sem solução	12	25.5%
TOTAL com objetivo disponível	35	74.5%

Legenda: OK=resolvido, T=time-limit, I=iter-limit, F=inválvel, E=erro numérico, M=sem conversão, X=crash, S=progresso lento.

3.3 Tabela detalhada por problema (Parte 1)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
arki0003	I 5119.4131 t=259.6s g=—	T — t=600.6s g=—	T — t=600.6s g=—	I 5119.4131 t=259.6s g=—	T -1440.0000 t=350.9s g=—	-	—	~ 5119.4131 (ipopt, TL)
arki0009	E — t=38.7s g=—	E — t=33.2s g=—	E — t=35.0s g=—	E — t=38.7s g=—	E — t=51.9s g=—	-	—	sem solução
bearing_400	OK -0.1380 t=52.5s g=1.87e-06	OK -0.1380 t=73.7s g=0.00e+00	E — t=40.8s g=—	OK -0.1380 t=52.5s g=1.87e-06	T 15.7282 t=412.5s g=—	madnlp	-0.1380	OK -0.1380 (madnlp)
camshape_6400	OK 4.4478 t=28.7s g=7.52e-11	OK 4.4478 t=47.3s g=0.00e+00	T — t=601.3s g=—	OK 4.4478 t=28.7s g=7.52e-11	T 4.7081 t=318.2s g=—	madnlp	4.4478	OK 4.4478 (madnlp)
clnlbeam	E — t=13.0s g=—	E — t=15.8s g=—	E — t=13.9s g=—	E — t=13.0s g=—	E — t=13.0s g=—	-	—	sem solução
cont5_1_I	OK 2.7210 t=47.8s g=0.00e+00	OK 2.7210 t=95.1s g=3.48e-09	E — t=20.8s g=—	OK 2.7210 t=47.8s g=0.00e+00	T — t=600.9s g=—	ipopt	2.7210	OK 2.7210 (ipopt)
cont5_2_1_I	OK 7.128e-04 t=52.7s g=0.00e+00	OK 7.128e-04 t=135.4s g=1.68e-08	E — t=18.6s g=—	OK 7.128e-04 t=52.7s g=0.00e+00	T — t=600.8s g=—	ipopt	7.128e-04	OK 7.128e-04 (ipopt)
cont5_2_2_I	OK 5.867e-04 t=79.2s g=6.93e-09	OK 5.867e-04 t=142.8s g=0.00e+00	E — t=21.2s g=—	OK 5.867e-04 t=79.2s g=6.93e-09	T — t=600.7s g=—	madnlp	5.867e-04	OK 5.867e-04 (madnlp)
cont5_2_3_I	T 8.281e-04 t=325.3s g=—	OK 8.281e-04 t=107.2s g=0.00e+00	E — t=22.2s g=—	T 8.281e-04 t=325.3s g=—	T — t=601.1s g=—	madnlp	8.281e-04	OK 8.281e-04 (madnlp)
cont5_2_4_I	OK 0.0664 t=43.8s g=0.00e+00	OK 0.0664 t=80.5s g=7.57e-07	E — t=20.2s g=—	OK 0.0664 t=43.8s g=0.00e+00	T — t=600.3s g=—	ipopt	0.0664	OK 0.0664 (ipopt)
corkscrw	OK 98.0964 t=96.8s g=7.34e-10	OK 98.0964 t=130.4s g=0.00e+00	E — t=16.3s g=—	OK 98.0964 t=96.8s g=7.34e-10	T — t=601.0s g=—	madnlp	98.0964	OK 98.0964 (madnlp)
dirichlet120	OK 0.1888 t=133.4s g=7.30e-11	OK 0.1888 t=530.4s g=0.00e+00	E — t=206.5s g=—	OK 0.1888 t=133.4s g=7.30e-11	T — t=600.6s g=—	madnlp	0.1888	OK 0.1888 (madnlp)
dtoc1nd	OK 136.4558 t=31.6s g=0.00e+00	OK 136.4558 t=81.1s g=4.17e-16	T — t=601.1s g=—	OK 136.4558 t=31.6s g=0.00e+00	T — t=600.4s g=—	ipopt	136.4558	OK 136.4558 (ipopt)
dtoc2	T — t=601.1s g=—	T — t=600.9s g=—	E — t=57.4s g=—	T — t=601.1s g=—	T — t=600.8s g=—	-	—	sem solução
elec_400	T 8.113e+04 t=326.4s g=—	T — t=601.1s g=—	T — t=600.6s g=—	T 8.113e+04 t=326.4s g=—	T 7.429e+04 t=562.7s g=—	-	—	~ 7.429e+04 (uno, TL)
ex1_160	T — t=600.7s g=—	T — t=600.3s g=—	T — t=600.2s g=—	T — t=600.7s g=—	T — t=600.8s g=—	-	—	sem solução
ex1_320	T — t=601.0s g=—	T — t=601.1s g=—	T — t=600.4s g=—	T — t=601.0s g=—	T — t=600.4s g=—	-	—	sem solução
ex2_160	T — t=600.4s g=—	T — t=600.1s g=—	T — t=601.0s g=—	T — t=600.4s g=—	T — t=—s g=—	-	—	sem solução
ex2_320	T — t=601.0s g=—	T — t=600.8s g=—	T — t=600.4s g=—	T — t=601.0s g=—	T — t=—s g=—	-	—	sem solução
ex3_160	T — t=600.2s g=—	T — t=600.4s g=—	T — t=600.4s g=—	T — t=600.2s g=—	T — t=—s g=—	-	—	sem solução
ex3_320	T — t=600.8s g=—	T — t=601.0s g=—	T — t=600.6s g=—	T — t=600.8s g=—	T — t=—s g=—	-	—	sem solução
ex4_160	OK 0.0847 t=24.0s g=4.34e-03	OK 0.0803 t=53.2s g=0.00e+00	E — t=20.8s g=—	OK 0.0847 t=24.0s g=4.34e-03	T — t=—s g=—	madnlp	0.0803	OK 0.0803 (madnlp)
ex4_320	OK 0.0889 t=52.9s g=5.15e-05	OK 0.0889 t=84.5s g=0.00e+00	E — t=30.8s g=—	OK 0.0889 t=52.9s g=5.15e-05	T — t=—s g=—	madnlp	0.0889	OK 0.0889 (madnlp)
ex5_160	OK 0.0627 t=24.7s g=2.15e-03	OK 0.0605 t=45.1s g=0.00e+00	E — t=26.5s g=—	OK 0.0627 t=24.7s g=2.15e-03	T — t=—s g=—	madnlp	0.0605	OK 0.0605 (madnlp)
ex5_320	OK 0.0826 t=55.2s g=6.35e-03	OK 0.0762 t=74.6s g=0.00e+00	E — t=38.0s g=—	OK 0.0826 t=55.2s g=6.35e-03	T — t=—s g=—	madnlp	0.0762	OK 0.0762 (madnlp)
ex6_160	OK -4.3053 t=25.0s g=0.00e+00	OK -4.3053 t=45.9s g=1.09e-05	E — t=17.3s g=—	OK -4.3053 t=25.0s g=0.00e+00	T — t=—s g=—	ipopt	-4.3053	OK -4.3053 (ipopt)
ex6_320	OK -4.3293 t=45.0s g=2.51e-07	OK -4.3293 t=90.1s g=0.00e+00	E — t=44.1s g=—	OK -4.3293 t=45.0s g=2.51e-07	T — t=—s g=—	madnlp	-4.3293	OK -4.3293 (madnlp)
ex8_2_2	E — t=17.6s g=—	E — t=21.5s g=—	E — t=22.2s g=—	E — t=17.6s g=—	T — t=—s g=—	-	—	sem solução
ex8_2_3	E — t=36.7s g=—	E — t=38.2s g=—	E — t=45.7s g=—	E — t=36.7s g=—	T — t=—s g=—	-	—	sem solução

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
gasoil_1600	OK 0.0052 t=28.5s g=0.00e+00	T — t=601.1s g=—	T — t=601.9s g=—	OK 0.0052 t=28.5s g=0.00e+00	- — t=—s g=—	ipopt	0.0052	OK 0.0052 (ipopt)
gasoil_3200	OK 0.0052 t=37.3s g=0.00e+00	T — t=600.6s g=—	E — t=57.6s g=—	OK 0.0052 t=37.3s g=0.00e+00	- — t=—s g=—	ipopt	0.0052	OK 0.0052 (ipopt)
henon120	T 8.2198 t=403.2s g=—	T — t=600.5s g=—	E — t=99.1s g=—	T 8.2198 t=403.2s g=—	- — t=—s g=—	-	—	~ 8.2198 (ipopt, TL)
lane_emden80	OK 1.9078 t=189.7s g=0.00e+00	OK 1.9078 t=178.6s g=6.98e-16	E — t=144.4s g=—	OK 1.9078 t=189.7s g=0.00e+00	- — t=—s g=—	ipopt	1.9078	OK 1.9078 (ipopt)
lukvie10	OK 8.839e+04 t=51.3s g=0.00e+00	OK 8.839e+04 t=53.0s g=0.00e+00	E — t=18.5s g=—	OK 8.839e+04 t=51.3s g=0.00e+00	- — t=—s g=—	ipopt	8.839e+04	OK 8.839e+04 (ipopt)
lukvie14	OK 1.902e+06 t=41.9s g=0.00e+00	OK 1.902e+06 t=45.6s g=0.00e+00	E — t=24.8s g=—	OK 1.902e+06 t=41.9s g=0.00e+00	- — t=—s g=—	ipopt	1.902e+06	OK 1.902e+06 (ipopt)
lukvli10	OK 8.839e+04 t=69.9s g=1.65e-16	OK 8.839e+04 t=65.8s g=0.00e+00	E — t=31.0s g=—	OK 8.839e+04 t=69.9s g=1.65e-16	- — t=—s g=—	madnlp	8.839e+04	OK 8.839e+04 (madnlp)
lukvli14	OK 1.902e+06 t=41.8s g=0.00e+00	T — t=601.0s g=—	E — t=23.6s g=—	OK 1.902e+06 t=41.8s g=0.00e+00	- — t=—s g=—	ipopt	1.902e+06	OK 1.902e+06 (ipopt)
marine_1600	OK 1.975e+07 t=30.1s g=0.00e+00	T — t=601.0s g=—	E — t=17.2s g=—	OK 1.975e+07 t=30.1s g=0.00e+00	- — t=—s g=—	ipopt	1.975e+07	OK 1.975e+07 (ipopt)
optmass	OK -0.1032 t=13.1s g=1.64e-02	OK -0.1195 t=28.8s g=0.00e+00	E — t=14.5s g=—	OK -0.1032 t=13.1s g=1.64e-02	- — t=—s g=—	madnlp	-0.1195	OK -0.1195 (madnlp)
pinene_1600	OK 19.8722 t=29.9s g=0.00e+00	OK 19.8722 t=40.8s g=1.61e-15	E — t=52.9s g=—	OK 19.8722 t=29.9s g=0.00e+00	- — t=—s g=—	ipopt	19.8722	OK 19.8722 (ipopt)
pinene_3200	OK 19.8722 t=39.0s g=5.36e-16	OK 19.8722 t=51.3s g=0.00e+00	E — t=20.2s g=—	OK 19.8722 t=39.0s g=5.36e-16	- — t=—s g=—	madnlp	19.8722	OK 19.8722 (madnlp)
qcqp1000-1c	T — t=600.7s g=—	T — t=601.2s g=—	- — t=—s g=—	T — t=600.7s g=—	- — t=—s g=—	-	—	sem solução
qcqp500-1c	OK -2.735e+06 t=392.0s g=4.92e-09	OK -2.735e+06 t=450.0s g=0.00e+00	- — t=—s g=—	OK -2.735e+06 t=392.0s g=4.92e-09	- — t=—s g=—	madnlp	-2.735e+06	OK -2.735e+06 (madnlp)
robot_1600	OK 9.1413 t=25.5s g=4.30e-10	OK 9.1413 t=38.2s g=0.00e+00	- — t=—s g=—	OK 9.1413 t=25.5s g=4.30e-10	- — t=—s g=—	madnlp	9.1413	OK 9.1413 (madnlp)
rocket_12800	T 1.1040 t=337.9s g=—	T — t=600.9s g=—	- — t=—s g=—	T 1.1040 t=337.9s g=—	- — t=—s g=—	-	—	~ 1.1040 (ipopt, TL)
steering_12800	OK 0.5546 t=36.2s g=1.87e-12	OK 0.5546 t=47.7s g=0.00e+00	- — t=—s g=—	OK 0.5546 t=36.2s g=1.87e-12	- — t=—s g=—	madnlp	0.5546	OK 0.5546 (madnlp)
svanberg	OK 8.362e+04 t=80.6s g=0.00e+00	OK 8.362e+04 t=54.4s g=6.86e-14	- — t=—s g=—	OK 8.362e+04 t=80.6s g=0.00e+00	- — t=—s g=—	ipopt	8.362e+04	OK 8.362e+04 (ipopt)

4. Parte 2 — COCONUT Library 1 (202 problemas)

4.1 Resumo por solver

Solver	Total	Resolv.	%	Gap $\leq 1e-4$	Gap $\leq 1e-6$	T.médio(s)	T.med.(s)
ipopt	202	190	94.1%	130	126	0.6	0.1
madnlp	202	192	95.0%	128	126	1.1	0.1
nlopt	202	188	93.1%	106	99	7.8	0.1
optim	202	151	74.8%	100	97	2.1	0.2
uno	202	138	68.3%	95	94	14.8	15.1

Distribuição do 'melhor solver': ipopt=43 | madnlp=51 | nlopt=32 | optim=47 | uno=27

4.2 Cobertura na melhor situação de solução

Situação (melhor caso)	Qtd	%
Resolvido (algum solver convergiu)	200	99.0%
Melhor incumbente (no time-limit)	2	1.0%
Sem solução	0	0.0%
TOTAL com objetivo disponível	202	100.0%

4.3 Tabela detalhada por problema (Parte 2)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
abel	OK 225.1946 t=5.6s g=8.00e-13	OK 225.1946 t=20.8s g=3.55e-13	OK 225.1946 t=9.0s g=1.54e-10	OK 225.1946 t=33.7s g=9.55e-11	OK 225.1946 t=9.1s g=0.00e+00	uno	225.1946	OK 225.1946 (uno)
alkyl	OK -1.7650 t=7.5s g=5.41e-12	OK -1.7650 t=21.6s g=0.00e+00	OK -1.7650 t=7.4s g=2.65e-07	OK -1.7650 t=2.9s g=3.00e-07	OK -1.7650 t=9.0s g=3.00e-07	madnlp	-1.7650	OK -1.7650 (madnlp)
camshape100	OK -4.2842 t=1.8s g=2.10e-01	OK -4.2842 t=4.8s g=2.10e-01	OK -3.3018 t=4.0s g=3.91e-01	OK -5.4258 t=8.0s g=0.00e+00	E -3.3380 t=20.5s g=—	optim	-5.4258	OK -5.4258 (optim)
camshape200	OK -4.2787 t=1.8s g=0.00e+00	OK -4.2787 t=1.7s g=5.38e-14	OK -3.3057 t=20.9s g=2.27e-01	E -6.0182 t=11.8s g=—	E -3.2199 t=30.1s g=—	ipopt	-4.2787	OK -4.2787 (ipopt)
catmix100	OK -0.0481 t=1.5s g=0.00e+00	OK -0.0481 t=1.3s g=6.73e-08	OK -0.0481 t=16.8s g=5.17e-06	OK -0.0437 t=3.6s g=4.40e-03	E -9.8940 t=1.7s g=—	ipopt	-0.0481	OK -0.0481 (ipopt)
catmix200	OK -0.0481 t=2.7s g=1.65e-02	OK -0.0481 t=2.6s g=1.65e-02	OK -0.0481 t=367.0s g=1.65e-02	OK -0.0645 t=4.6s g=0.00e+00	E -10.4125 t=3.9s g=—	optim	-0.0645	OK -0.0645 (optim)
chain100	OK 5.0698 t=1.0s g=5.05e+00	OK 5.0698 t=2.4s g=5.05e+00	OK 0.0200 t=1.1s g=0.00e+00	OK 5.0700 t=1.4s g=5.05e+00	F 8.6836 t=2.0s g=—	nlopt	0.0200	OK 0.0200 (nlopt)
chain200	OK 5.0689 t=3.0s g=5.06e+00	OK 5.0689 t=1.7s g=5.06e+00	OK 0.0100 t=1.6s g=0.00e+00	E 5.0725 t=12.0s g=—	OK 5.0689 t=7.3s g=5.06e+00	nlopt	0.0100	OK 0.0100 (nlopt)
chain400	OK 5.0686 t=3.4s g=5.06e+00	OK 5.0686 t=4.6s g=5.06e+00	OK 0.0050 t=4.7s g=0.00e+00	E 5.0840 t=13.7s g=—	F 21.9796 t=6.0s g=—	nlopt	0.0050	OK 0.0050 (nlopt)
chain50	OK 5.0723 t=0.4s g=5.03e+00	OK 5.0723 t=0.4s g=5.03e+00	OK 0.0400 t=0.4s g=0.00e+00	E 5.0755 t=2.1s g=—	F 7.3657 t=0.4s g=—	nlopt	0.0400	OK 0.0400 (nlopt)
chakra	OK -179.1336 t=0.3s g=1.65e-10	OK -179.1336 t=0.4s g=1.65e-10	OK -179.1331 t=0.3s g=2.74e-06	OK -163.7487 t=0.3s g=8.59e-02	OK -179.1336 t=0.3s g=0.00e+00	uno	-179.1336	OK -179.1336 (uno)
chance	OK 29.8944 t=0.8s g=0.00e+00	OK 29.8944 t=1.1s g=7.97e-13	I 0.0000 t=3.2s g=—	OK 29.8944 t=1.4s g=9.39e-10	E 0.0000 t=1.0s g=—	ipopt	29.8944	OK 29.8944 (ipopt)
chem	OK -47.7065 t=0.2s g=0.00e+00	OK -47.7065 t=0.2s g=0.00e+00	OK -47.7065 t=0.2s g=1.89e-09	E -175.6688 t=0.3s g=—	OK -47.7065 t=0.2s g=2.52e-11	madnlp	-47.7065	OK -47.7065 (madnlp)
chenery	OK -1058.9199 t=0.4s g=0.00e+00	OK -1058.9199 t=0.4s g=0.00e+00	OK 0.0000 t=0.3s g=1.00e+00	E — t=2.2s g=—	OK -1058.9199 t=0.7s g=3.19e-10	madnlp	-1058.9199	OK -1058.9199 (madnlp)
circle	OK 4.5742 t=0.4s g=4.57e+00	OK 4.5742 t=0.4s g=4.57e+00	OK 7.772e-15 t=0.8s g=0.00e+00	OK 5.743e-04 t=0.2s g=5.74e-04	F 0.0000 t=0.4s g=—	nlopt	7.772e-15	OK 7.772e-15 (nlopt)
demo7	OK -1.589e+06 t=0.3s g=3.78e-10	OK -1.589e+06 t=0.5s g=0.00e+00	OK -1.589e+06 t=0.9s g=2.83e-09	OK 1593.1974 t=0.8s g=1.00e+00	F -1.004e+05 t=0.6s g=—	madnlp	-1.589e+06	OK -1.589e+06 (madnlp)
dispatch	OK 3155.2879 t=0.0s g=0.00e+00	OK 3155.2879 t=0.0s g=0.00e+00	OK 3155.2879 t=0.0s g=8.69e-09	OK 3155.2879 t=0.0s g=8.69e-09	X — t=25.4s g=—	madnlp	3155.2879	OK 3155.2879 (madnlp)
elec25	OK 243.8128 t=1.4s g=5.27e-01	OK 243.8128 t=36.9s g=5.27e-01	OK 243.8130 t=16.5s g=5.27e-01	OK 159.7174 t=38.3s g=0.00e+00	T 177.9126 t=43.0s g=—	optim	159.7174	OK 159.7174 (optim)
elec50	OK 1055.1823 t=8.3s g=2.29e-01	OK 1055.1823 t=9.9s g=2.29e-01	OK 1055.1831 t=7.3s g=2.29e-01	OK 858.6808 t=7.0s g=0.00e+00	T 843.8928 t=46.7s g=—	optim	858.6808	OK 858.6808 (optim)
etamac	OK -15.2947 t=0.4s g=3.34e-14	OK -15.2947 t=0.5s g=0.00e+00	OK -15.2940 t=0.8s g=4.25e-05	OK -15.2937 t=0.6s g=6.58e-05	OK -15.2947 t=21.1s g=1.36e-11	madnlp	-15.2947	OK -15.2947 (madnlp)
ex14_1_1	OK 2.361e-11 t=0.7s g=7.84e-01	OK 2.361e-11 t=0.8s g=7.84e-01	OK -6.661e-16 t=0.8s g=7.84e-01	OK -0.7841 t=0.8s g=0.00e+00	OK -1.943e-16 t=16.4s g=7.84e-01	optim	-0.7841	OK -0.7841 (optim)
ex14_1_2	OK 1.004e-08 t=0.1s g=1.00e-08	OK 1.004e-08 t=0.1s g=1.00e-08	OK -2.313e-12 t=0.1s g=0.00e+00	OK 5.854e-04 t=0.2s g=5.85e-04	I 0.0231 t=18.5s g=—	nlopt	-2.313e-12	OK -2.313e-12 (nlopt)
ex14_1_3	OK 2.361e-11 t=0.1s g=2.36e-11	OK 2.361e-11 t=0.1s g=2.36e-11	OK 7.670e-14 t=0.1s g=7.66e-14	OK 1.110e-16 t=0.1s g=5.51e-17	OK 5.596e-17 t=18.5s g=0.00e+00	uno	5.596e-17	OK 5.596e-17 (uno)
ex14_1_4	OK 4.298e-10 t=0.1s g=4.30e-10	OK 9.131e-10 t=0.0s g=9.13e-10	OK 2.228e-15 t=0.0s g=2.23e-15	OK 4.929e-16 t=0.1s g=4.93e-16	OK 0.0000 t=16.6s g=0.00e+00	uno	0.0000	OK 0.0000 (uno)
ex14_1_5	OK -4.982e-09 t=0.1s g=5.75e-12	OK -4.988e-09 t=0.1s g=0.00e+00	OK -6.395e-15 t=0.1s g=4.99e-09	OK -3.018e-17 t=0.1s g=4.99e-09	OK 2.511e-17 t=16.3s g=4.99e-09	madnlp	-4.988e-09	OK -4.988e-09 (madnlp)
ex14_1_6	OK 1.0000 t=0.3s g=2.22e-16	OK 1.0000 t=0.2s g=0.00e+00	OK 1.0000 t=0.1s g=7.49e-09	E — t=0.8s g=—	OK 1.0000 t=9.3s g=7.49e-09	madnlp	1.0000	OK 1.0000 (madnlp)
ex14_1_7	OK 0.1350 t=0.2s g=1.35e-01	OK 3.012e-08 t=0.2s g=3.01e-08	OK 0.0000 t=0.2s g=0.00e+00	OK 165.8401 t=0.3s g=1.66e+02	OK 7.0427 t=16.1s g=7.04e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex14_1_8	OK 0.0414 t=0.1s g=2.99e-02	OK 0.0414 t=0.1s g=2.99e-02	OK 0.0414 t=0.1s g=2.99e-02	OK 0.0116 t=0.2s g=0.00e+00	OK 0.0414 t=15.3s g=2.99e-02	optim	0.0116	OK 0.0116 (optim)
ex14_1_9	OK -4.984e-09 t=0.0s g=2.54e-13	OK -4.985e-09 t=0.0s g=0.00e+00	OK 1.414e-14 t=0.0s g=4.98e-09	OK 5.156e-16 t=0.1s g=4.98e-09	OK -8.674e-19 t=16.9s g=4.98e-09	madnlp	-4.985e-09	OK -4.985e-09 (madnlp)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
ex14_2_1	OK 7.518e-09 t=0.1s g=7.52e-09	OK 7.511e-09 t=0.1s g=7.51e-09	OK 0.0000 t=0.1s g=0.00e+00	OK 2.796e-16 t=0.2s g=2.80e-16	OK 0.0000 t=18.4s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex14_2_2	OK 2.530e-09 t=0.0s g=2.53e-09	OK 2.530e-09 t=0.0s g=2.53e-09	OK 0.0000 t=0.0s g=0.00e+00	OK 8.882e-16 t=0.1s g=8.88e-16	OK 0.0000 t=18.0s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex14_2_3	OK 1.267e-08 t=0.1s g=1.27e-08	OK 1.299e-08 t=0.1s g=1.30e-08	OK 0.0000 t=0.1s g=0.00e+00	OK 8.882e-16 t=0.2s g=8.88e-16	OK 0.0000 t=15.7s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex14_2_4	OK 7.564e-09 t=0.1s g=7.56e-09	OK 7.541e-09 t=0.1s g=7.54e-09	OK 2.080e-14 t=0.1s g=2.08e-14	OK 8.882e-16 t=0.1s g=8.88e-16	OK 0.0000 t=15.3s g=0.00e+00	uno	0.0000	OK 0.0000 (uno)
ex14_2_5	OK 2.532e-09 t=0.0s g=2.53e-09	OK 2.533e-09 t=0.0s g=2.53e-09	OK 0.0000 t=0.0s g=0.00e+00	OK 8.882e-16 t=0.1s g=8.88e-16	OK 0.0000 t=15.2s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex14_2_6	OK 8.458e-09 t=0.1s g=8.46e-09	OK 7.980e-09 t=0.1s g=7.98e-09	OK 0.0000 t=0.1s g=0.00e+00	OK 3.996e-20 t=0.2s g=4.00e-20	OK 0.0000 t=15.2s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex14_2_7	OK 1.258e-08 t=0.1s g=1.26e-08	OK 1.258e-08 t=0.2s g=1.26e-08	OK 1.189e-04 t=0.1s g=1.19e-04	OK 8.882e-16 t=0.2s g=8.88e-16	OK 0.0000 t=15.1s g=0.00e+00	uno	0.0000	OK 0.0000 (uno)
ex14_2_8	OK 2.693e-09 t=0.1s g=2.69e-09	OK 2.623e-09 t=0.0s g=2.62e-09	OK 6.661e-16 t=0.1s g=6.66e-16	OK 8.882e-16 t=0.1s g=8.88e-16	OK 0.0000 t=14.9s g=0.00e+00	uno	0.0000	OK 0.0000 (uno)
ex14_2_9	OK 2.530e-09 t=0.1s g=2.53e-09	OK 2.530e-09 t=0.1s g=2.53e-09	OK 2.424e-14 t=0.1s g=2.42e-14	OK 3.423e-16 t=0.1s g=3.42e-16	OK 0.0000 t=15.8s g=0.00e+00	uno	0.0000	OK 0.0000 (uno)
ex2_1_1	OK -2.242e-06 t=0.0s g=1.00e+00	OK -2.242e-06 t=0.0s g=1.00e+00	OK 0.0000 t=0.0s g=1.00e+00	OK -16.5000 t=0.0s g=0.00e+00	OK 0.0000 t=9.5s g=1.00e+00	optim	-16.5000	OK -16.5000 (optim)
ex2_1_10	OK 4.932e+04 t=0.1s g=0.00e+00	OK 4.932e+04 t=0.1s g=5.16e-15	E 5.477e+05 t=0.1s g=—	OK 4.932e+04 t=0.2s g=1.61e-07	OK 4.932e+04 t=8.8s g=1.61e-07	ipopt	4.932e+04	OK 4.932e+04 (ipopt)
ex2_1_2	OK -213.0000 t=0.0s g=0.00e+00	OK -213.0000 t=0.0s g=0.00e+00	OK -213.0000 t=0.0s g=1.87e-08	OK -213.0000 t=0.1s g=1.87e-08	OK -213.0000 t=9.0s g=1.87e-08	ipopt	-213.0000	OK -213.0000 (ipopt)
ex2_1_3	OK -10.1094 t=0.1s g=3.26e-01	OK -10.1094 t=0.1s g=3.26e-01	OK -10.1094 t=0.1s g=3.26e-01	OK -15.0000 t=0.1s g=0.00e+00	OK -10.1094 t=10.9s g=3.26e-01	optim	-15.0000	OK -15.0000 (optim)
ex2_1_4	OK -11.0000 t=0.0s g=8.33e-02	OK -11.0000 t=0.1s g=8.33e-02	OK -11.0000 t=0.1s g=8.33e-02	OK -11.9999 t=0.1s g=0.00e+00	OK -11.0000 t=10.3s g=8.33e-02	optim	-11.9999	OK -11.9999 (optim)
ex2_1_5	OK -268.0146 t=0.1s g=1.83e-01	OK -268.0146 t=0.1s g=1.83e-01	OK -268.0146 t=0.1s g=1.83e-01	OK -328.0216 t=0.2s g=0.00e+00	OK -268.0146 t=9.9s g=1.83e-01	optim	-328.0216	OK -328.0216 (optim)
ex2_1_6	OK -29.0000 t=0.1s g=3.96e-01	OK -29.0000 t=0.1s g=3.96e-01	OK -9.7180 t=0.1s g=7.98e-01	OK -47.9981 t=0.1s g=0.00e+00	OK -9.7180 t=8.8s g=7.98e-01	optim	-47.9981	OK -47.9981 (optim)
ex2_1_7	OK -407.2493 t=0.1s g=7.84e-11	OK -407.2493 t=0.1s g=0.00e+00	OK -407.2493 t=0.1s g=9.07e-09	OK -407.1034 t=0.2s g=3.58e-04	OK -407.2493 t=11.9s g=9.17e-09	madnlp	-407.2493	OK -407.2493 (madnlp)
ex2_1_8	OK 1.599e+04 t=0.2s g=1.04e+00	OK 4.348e+04 t=0.1s g=1.11e+00	OK 1.828e+04 t=0.1s g=1.04e+00	OK -4.089e+05 t=0.1s g=0.00e+00	OK 1.599e+04 t=9.3s g=1.04e+00	optim	-4.089e+05	OK -4.089e+05 (optim)
ex2_1_9	OK -0.3750 t=0.1s g=0.00e+00	OK -0.3750 t=0.0s g=6.88e-15	OK -0.3750 t=0.0s g=1.22e-08	E -29.6296 t=0.1s g=—	OK -0.3333 t=10.1s g=4.17e-02	ipopt	-0.3750	OK -0.3750 (ipopt)
ex3_1_1	OK 7049.2479 t=0.1s g=1.85e-08	OK 7049.2478 t=0.1s g=0.00e+00	OK 7049.2480 t=0.0s g=3.63e-08	E — t=0.5s g=—	I 3410.9617 t=11.3s g=—	madnlp	7049.2478	OK 7049.2478 (madnlp)
ex3_1_2	OK -3.067e+04 t=0.1s g=0.00e+00	OK -3.067e+04 t=0.1s g=2.28e-14	OK -3.067e+04 t=0.0s g=5.54e-09	OK -3.067e+04 t=0.1s g=6.24e-09	I -3.067e+04 t=9.0s g=—	ipopt	-3.067e+04	OK -3.067e+04 (ipopt)
ex3_1_3	OK -132.0000 t=0.4s g=6.15e-12	OK -132.0000 t=0.3s g=0.00e+00	OK -132.0000 t=0.5s g=8.72e-09	E — t=1.0s g=—	OK -132.0000 t=8.7s g=8.72e-09	madnlp	-132.0000	OK -132.0000 (madnlp)
ex3_1_4	OK -4.0000 t=0.0s g=0.00e+00	OK -4.0000 t=0.0s g=3.81e-11	OK -4.0000 t=0.0s g=1.82e-08	OK -4.0000 t=0.0s g=1.82e-08	E -6.0000 t=8.5s g=—	ipopt	-4.0000	OK -4.0000 (ipopt)
ex4_1_1	OK -0.5200 t=0.0s g=0.00e+00	OK -0.5200 t=0.0s g=2.22e-16	OK -0.5200 t=0.0s g=2.22e-16	OK -0.5200 t=6.3s g=1.11e-16	OK -0.5200 t=15.0s g=1.11e-16	ipopt	-0.5200	OK -0.5200 (ipopt)
ex4_1_2	OK -663.5001 t=0.4s g=6.85e-16	OK -663.5001 t=0.4s g=0.00e+00	OK -663.5001 t=0.4s g=1.54e-14	OK -663.5001 t=0.4s g=1.71e-16	E -663.5001 t=14.6s g=—	madnlp	-663.5001	OK -663.5001 (madnlp)
ex4_1_3	OK -3.878e-07 t=0.0s g=1.00e+00	OK -3.878e-07 t=0.0s g=1.00e+00	OK 0.0000 t=0.0s g=1.00e+00	OK -443.6717 t=0.0s g=0.00e+00	OK 0.0000 t=13.9s g=1.00e+00	optim	-443.6717	OK -443.6717 (optim)
ex4_1_4	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=15.3s g=0.00e+00	madnlp	0.0000	OK 0.0000 (madnlp)
ex4_1_5	OK 7.176e-20 t=0.1s g=7.18e-20	OK 7.176e-20 t=0.2s g=7.18e-20	OK 0.0000 t=0.2s g=0.00e+00	OK 0.0000 t=0.1s g=0.00e+00	OK 0.0000 t=15.3s g=0.00e+00	optim	0.0000	OK 0.0000 (optim)
ex4_1_6	OK 250.0000 t=0.0s g=0.00e+00	OK 250.0000 t=0.0s g=0.00e+00	OK 250.0000 t=0.0s g=0.00e+00	OK 250.0000 t=0.0s g=0.00e+00	OK 250.0000 t=14.9s g=0.00e+00	optim	250.0000	OK 250.0000 (optim)
ex4_1_7	OK -7.5000 t=0.0s g=1.18e-16	OK -7.5000 t=0.0s g=1.18e-16	OK -7.5000 t=0.0s g=1.18e-16	OK -7.5000 t=0.0s g=0.00e+00	OK -7.5000 t=14.4s g=2.37e-16	optim	-7.5000	OK -7.5000 (optim)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
ex4_1_8	OK -16.7389 t=0.0s g=3.94e-09	OK -16.7389 t=0.0s g=3.94e-09	OK -16.7389 t=0.0s g=0.00e+00	OK -16.7389 t=0.0s g=3.94e-09	OK -16.7389 t=14.9s g=3.94e-09	nlopt	-16.7389	OK -16.7389 (nlopt)
ex4_1_9	OK -4.0537 t=0.0s g=8.29e-02	OK -4.0537 t=0.0s g=8.29e-02	OK -4.0537 t=0.0s g=8.29e-02	OK -4.4200 t=0.0s g=0.00e+00	OK -4.0537 t=14.3s g=8.29e-02	optim	-4.4200	OK -4.4200 (optim)
ex5_2_2_case1	OK -400.0000 t=0.1s g=0.00e+00	OK -100.0000 t=0.1s g=7.50e-01	OK 0.0000 t=0.1s g=1.00e+00	OK -400.0000 t=0.1s g=1.03e-08	OK 0.0000 t=9.7s g=1.00e+00	ipopt	-400.0000	OK -400.0000 (ipopt)
ex5_2_2_case2	OK -400.0000 t=0.1s g=3.33e-01	OK -400.0000 t=0.1s g=3.33e-01	OK 0.0000 t=0.0s g=1.00e+00	OK -600.0000 t=0.1s g=0.00e+00	OK 0.0000 t=8.5s g=1.00e+00	optim	-600.0000	OK -600.0000 (optim)
ex5_2_2_case3	OK -750.0000 t=0.1s g=0.00e+00	OK -750.0000 t=0.1s g=2.54e-12	OK 0.0000 t=0.0s g=1.00e+00	OK -750.0000 t=0.1s g=1.01e-08	OK 0.0000 t=9.6s g=1.00e+00	ipopt	-750.0000	OK -750.0000 (ipopt)
ex5_2_4	OK -450.0000 t=0.1s g=0.00e+00	OK -450.0000 t=0.1s g=1.47e-14	OK -255.9877 t=0.0s g=4.31e-01	E — t=0.4s g=—	OK -450.0000 t=9.6s g=1.47e-08	ipopt	-450.0000	OK -450.0000 (ipopt)
ex5_2_5	OK -3500.0000 t=0.3s g=2.90e-01	OK -3500.0000 t=0.5s g=2.90e-01	OK -3500.0013 t=0.2s g=2.90e-01	OK -4931.0899 t=0.2s g=0.00e+00	I -3500.0000 t=9.3s g=—	optim	-4931.0899	OK -4931.0899 (optim)
ex5_3_2	OK 1.8642 t=0.1s g=8.66e-01	OK 1.8642 t=0.1s g=8.66e-01	OK 0.9979 t=0.1s g=0.00e+00	OK 3.5176 t=0.1s g=2.52e+00	OK 1.8642 t=8.7s g=8.66e-01	nlopt	0.9979	OK 0.9979 (nlopt)
ex5_3_3	OK 3.2340 t=0.4s g=9.16e-01	OK 3.6603 t=0.4s g=1.17e+00	OK 1.6878 t=0.3s g=0.00e+00	E 6.5869 t=0.9s g=—	OK 3.2340 t=8.8s g=9.16e-01	nlopt	1.6878	OK 1.6878 (nlopt)
ex5_4_2	OK 7512.2300 t=0.1s g=1.72e-08	OK 7512.2299 t=0.0s g=0.00e+00	OK 7512.2301 t=0.0s g=3.43e-08	E — t=0.3s g=—	I 1.098e+04 t=8.4s g=—	madnlp	7512.2299	OK 7512.2299 (madnlp)
ex5_4_3	OK 5937.4370 t=0.1s g=1.08e+00	OK 4845.4620 t=0.1s g=6.98e-01	OK 4.562e+04 t=0.1s g=1.50e+01	OK 2853.4541 t=0.1s g=0.00e+00	OK 4845.4620 t=14.3s g=6.98e-01	optim	2853.4541	OK 2853.4541 (optim)
ex5_4_4	OK 1.184e+04 t=0.1s g=1.41e+00	OK 1.101e+04 t=0.1s g=1.24e+00	OK 2.891e+04 t=0.1s g=4.88e+00	OK 4917.2935 t=0.1s g=0.00e+00	OK 1.148e+04 t=14.5s g=1.33e+00	optim	4917.2935	OK 4917.2935 (optim)
ex6_1_1	OK -0.0176 t=0.1s g=2.33e-06	OK -0.0176 t=0.0s g=2.33e-06	OK -0.0021 t=0.1s g=1.55e-02	OK -0.0176 t=0.1s g=0.00e+00	T — t=20.3s g=—	optim	-0.0176	OK -0.0176 (optim)
ex6_1_2	OK 2.457e-06 t=0.1s g=3.25e-02	OK -0.0325 t=0.0s g=0.00e+00	OK 0.0274 t=0.0s g=5.98e-02	OK 2.595e-06 t=0.0s g=3.25e-02	E -0.0325 t=16.3s g=—	madnlp	-0.0325	OK -0.0325 (madnlp)
ex6_1_3	OK -0.3243 t=0.1s g=2.77e-02	OK -0.3243 t=0.1s g=2.77e-02	OK -0.3520 t=0.1s g=0.00e+00	OK -0.3245 t=0.1s g=2.76e-02	E -0.3270 t=16.5s g=—	nlopt	-0.3520	OK -0.3520 (nlopt)
ex6_1_4	OK 4.627e-07 t=0.0s g=2.95e-01	OK -0.2945 t=0.0s g=0.00e+00	OK -0.1128 t=0.0s g=1.82e-01	OK 1.7681 t=0.0s g=2.06e+00	I 0.0027 t=15.0s g=—	madnlp	-0.2945	OK -0.2945 (madnlp)
ex6_2_10	OK -2.6606 t=0.1s g=3.76e-10	OK -2.6606 t=0.1s g=3.76e-10	OK -2.6606 t=0.1s g=3.76e-10	OK -2.6606 t=0.1s g=0.00e+00	OK -2.6606 t=15.5s g=3.76e-10	optim	-2.6606	OK -2.6606 (optim)
ex6_2_11	OK 1.557e-06 t=0.0s g=4.44e-15	OK 1.557e-06 t=0.0s g=8.88e-16	OK 1.557e-06 t=0.0s g=0.00e+00	E 0.0233 t=0.1s g=—	OK 1.557e-06 t=14.6s g=1.33e-15	nlopt	1.557e-06	OK 1.557e-06 (nlopt)
ex6_2_12	OK 0.3936 t=0.0s g=0.00e+00	OK 0.3936 t=0.0s g=0.00e+00	OK 0.3936 t=0.0s g=1.28e-15	OK 0.3936 t=0.0s g=1.30e-10	OK 0.3936 t=14.9s g=1.89e-15	ipopt	0.3936	OK 0.3936 (ipopt)
ex6_2_13	OK -0.2162 t=0.0s g=1.08e-15	OK -0.2162 t=0.1s g=1.58e-15	OK -0.2162 t=0.1s g=0.00e+00	OK -0.2162 t=0.1s g=2.08e-15	OK -0.2162 t=15.7s g=1.39e-15	nlopt	-0.2162	OK -0.2162 (nlopt)
ex6_2_14	OK 0.0987 t=0.0s g=7.94e-01	OK 0.0987 t=0.0s g=7.94e-01	OK 0.0987 t=0.0s g=7.94e-01	OK 0.0987 t=0.0s g=7.94e-01	OK -0.6954 t=15.7s g=0.00e+00	uno	-0.6954	OK -0.6954 (uno)
ex6_2_5	OK -70.5581 t=0.1s g=1.53e-14	OK -70.5581 t=0.1s g=2.09e-14	OK -70.5581 t=0.1s g=2.27e-10	E -105.1380 t=0.2s g=—	OK -70.5581 t=14.4s g=0.00e+00	uno	-70.5581	OK -70.5581 (uno)
ex6_2_6	OK 7.108e-07 t=0.0s g=3.31e-06	OK 7.108e-07 t=0.0s g=3.31e-06	OK 7.108e-07 t=0.0s g=3.31e-06	E 0.0021 t=0.1s g=—	OK -2.603e-06 t=15.4s g=0.00e+00	uno	-2.603e-06	OK -2.603e-06 (uno)
ex6_2_7	OK -0.1128 t=0.1s g=4.80e-02	OK -0.1128 t=0.1s g=4.80e-02	OK -0.1128 t=0.1s g=4.80e-02	E -0.1611 t=0.3s g=—	OK -0.1608 t=14.7s g=0.00e+00	uno	-0.1608	OK -0.1608 (uno)
ex6_2_8	OK -1.444e-06 t=0.0s g=0.00e+00	OK -1.444e-06 t=0.0s g=1.01e-14	OK -1.444e-06 t=0.0s g=2.64e-15	E -0.0298 t=0.1s g=—	OK -1.444e-06 t=15.0s g=8.71e-15	ipopt	-1.444e-06	OK -1.444e-06 (ipopt)
ex6_2_9	OK 0.1565 t=0.0s g=1.91e-01	OK 0.1565 t=0.0s g=1.91e-01	OK 0.1565 t=0.0s g=1.91e-01	OK -0.0341 t=0.1s g=0.00e+00	OK 0.1565 t=15.1s g=1.91e-01	optim	-0.0341	OK -0.0341 (optim)
ex7_2_1	OK 1227.2254 t=0.2s g=2.08e-01	OK 1227.2254 t=0.2s g=2.08e-01	OK 1227.2260 t=0.2s g=2.08e-01	OK 1015.9309 t=0.2s g=0.00e+00	X — t=32.4s g=—	optim	1015.9309	OK 1015.9309 (optim)
ex7_2_10	OK 0.1000 t=0.3s g=2.50e-07	OK 0.1000 t=0.3s g=2.17e-08	OK 0.1000 t=0.3s g=0.00e+00	OK 0.1000 t=0.3s g=0.00e+00	OK 0.1000 t=17.1s g=0.00e+00	nlopt	0.1000	OK 0.1000 (nlopt)
ex7_2_2	OK -0.3888 t=0.0s g=8.67e-02	OK -0.3888 t=0.0s g=8.67e-02	OK -0.3888 t=0.0s g=8.67e-02	OK -0.4755 t=0.1s g=0.00e+00	X — t=31.4s g=—	optim	-0.4755	OK -0.4755 (optim)
ex7_2_3	OK 7049.2476 t=0.1s g=1.36e-12	OK 7049.2476 t=0.1s g=0.00e+00	OK 7049.2492 t=0.1s g=2.26e-07	E — t=0.4s g=—	OK 7049.2480 t=15.4s g=5.29e-08	madnlp	7049.2476	OK 7049.2476 (madnlp)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
ex7_2_4	OK 3.9180 t=0.1s g=4.70e+00	OK 3.9180 t=0.1s g=4.70e+00	OK 4.2029 t=0.0s g=4.99e+00	OK -0.7824 t=0.1s g=0.00e+00	OK 3.9180 t=15.8s g=4.70e+00	optim	-0.7824	OK -0.7824 (optim)
ex7_2_5	OK 1.012e+04 t=0.0s g=0.00e+00	OK 1.012e+04 t=0.1s g=2.37e-14	OK 1.012e+04 t=0.0s g=1.34e-08	OK 1.012e+04 t=0.1s g=1.45e-08	OK 1.012e+04 t=16.6s g=1.45e-08	ipopt	1.012e+04	OK 1.012e+04 (ipopt)
ex7_2_6	OK -83.2497 t=0.0s g=1.64e-01	OK -83.2497 t=0.0s g=1.64e-01	OK -83.2497 t=0.0s g=1.64e-01	OK -99.5500 t=0.0s g=0.00e+00	OK -83.2497 t=16.8s g=1.64e-01	optim	-99.5500	OK -99.5500 (optim)
ex7_2_7	OK -5.7398 t=0.0s g=4.02e-01	OK -5.7398 t=0.0s g=4.02e-01	OK -5.7398 t=0.0s g=4.02e-01	OK -9.5999 t=0.1s g=0.00e+00	OK -5.7398 t=16.7s g=4.02e-01	optim	-9.5999	OK -9.5999 (optim)
ex7_2_8	OK -6.0820 t=0.0s g=0.00e+00	OK -5.8169 t=0.0s g=4.36e-02	OK -5.8169 t=0.0s g=4.36e-02	OK -6.0820 t=0.1s g=1.53e-08	OK -5.8169 t=18.0s g=4.36e-02	ipopt	-6.0820	OK -6.0820 (ipopt)
ex7_2_9	OK 1.1436 t=0.1s g=0.00e+00	OK 1.1436 t=0.1s g=3.67e-07	OK 1.1436 t=0.1s g=4.09e-07	OK 1.1436 t=0.1s g=4.11e-07	OK 1.1436 t=21.3s g=4.11e-07	ipopt	1.1436	OK 1.1436 (ipopt)
ex7_3_1	OK 0.3417 t=0.1s g=0.00e+00	OK 0.3417 t=0.1s g=6.90e-11	OK 0.3417 t=0.0s g=6.17e-09	OK 0.3417 t=0.1s g=6.17e-09	OK 0.3417 t=19.1s g=6.17e-09	ipopt	0.3417	OK 0.3417 (ipopt)
ex7_3_2	OK 1.0899 t=0.1s g=4.07e-16	OK 1.0899 t=0.1s g=0.00e+00	OK 1.0899 t=0.1s g=4.82e-08	E — t=0.3s g=—	OK 1.0899 t=19.3s g=4.82e-08	madnlp	1.0899	OK 1.0899 (madnlp)
ex7_3_3	OK 1.1448 t=0.0s g=4.18e-09	OK 1.1448 t=0.0s g=0.00e+00	OK 1.1448 t=0.0s g=4.62e-08	OK 36.1126 t=0.1s g=3.05e+01	X — t=25.3s g=—	madnlp	1.1448	OK 1.1448 (madnlp)
ex7_3_4	E 0.0000 t=0.1s g=—	OK 10.0000 t=0.2s g=2.03e+00	OK 0.0000 t=0.1s g=1.00e+00	OK -9.7547 t=0.1s g=0.00e+00	OK 10.0000 t=20.8s g=2.03e+00	optim	-9.7547	OK -9.7547 (optim)
ex7_3_5	E 0.0000 t=0.1s g=—	OK 1.2069 t=0.1s g=1.96e+00	OK 0.0000 t=0.1s g=7.56e-01	OK -0.7559 t=0.1s g=0.00e+00	OK 1.2069 t=19.5s g=1.96e+00	optim	-0.7559	OK -0.7559 (optim)
ex7_3_6	F -3.064e-09 t=0.3s g=—	F -3.011e-09 t=0.3s g=—	OK 0.0000 t=0.2s g=0.00e+00	E 0.5156 t=0.5s g=—	F 0.0000 t=18.8s g=—	nlopt	0.0000	OK 0.0000 (nlopt)
ex8_1_1	OK -2.0218 t=0.1s g=1.10e-15	OK -2.0218 t=0.0s g=0.00e+00	OK -2.0218 t=0.1s g=9.50e-09	OK -2.0218 t=0.1s g=9.78e-09	OK -2.0218 t=18.5s g=9.49e-09	madnlp	-2.0218	OK -2.0218 (madnlp)
ex8_1_2	OK -0.7970 t=0.0s g=2.56e-01	OK -0.7970 t=0.0s g=2.56e-01	OK 2.8035 t=0.0s g=3.62e+00	OK -1.0709 t=0.0s g=0.00e+00	OK 2.8035 t=17.2s g=3.62e+00	optim	-1.0709	OK -1.0709 (optim)
ex8_1_3	OK 30.0000 t=0.0s g=2.37e-16	OK 30.0000 t=0.0s g=2.37e-16	OK 30.0000 t=0.0s g=4.28e-10	OK 30.0000 t=1.0s g=1.18e-16	OK 30.0000 t=18.7s g=0.00e+00	uno	30.0000	OK 30.0000 (uno)
ex8_1_4	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=17.8s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
ex8_1_5	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=0.0s g=0.00e+00	OK 0.0000 t=16.2s g=0.00e+00	ipopt	0.0000	OK 0.0000 (ipopt)
ex8_1_6	OK -5.0654 t=0.0s g=4.98e-01	OK -5.0654 t=0.0s g=4.98e-01	OK -5.0413 t=0.0s g=5.00e-01	OK -5.0654 t=0.0s g=4.98e-01	OK -10.0860 t=14.8s g=0.00e+00	uno	-10.0860	OK -10.0860 (uno)
ex8_1_7	OK 0.0293 t=0.1s g=0.00e+00	OK 0.0293 t=0.1s g=9.57e-10	OK 1.0000 t=0.1s g=9.71e-01	OK 0.5698 t=0.1s g=5.40e-01	F 41.5047 t=17.6s g=—	ipopt	0.0293	OK 0.0293 (ipopt)
ex8_1_8	OK -0.3888 t=0.0s g=8.67e-02	OK -0.3888 t=0.0s g=8.67e-02	OK -0.3888 t=0.0s g=8.67e-02	OK -0.4755 t=0.1s g=0.00e+00	E -0.3860 t=15.6s g=—	optim	-0.4755	OK -0.4755 (optim)
ex8_2_1	OK -979.1783 t=0.3s g=0.00e+00	OK -979.1783 t=0.3s g=6.24e-13	OK -978.9757 t=0.3s g=2.07e-04	E — t=2.5s g=—	OK -979.1783 t=16.5s g=2.36e-08	ipopt	-979.1783	OK -979.1783 (ipopt)
ex8_2_4	E 2320.9985 t=0.7s g=—	E 0.0216 t=5.2s g=—	I -384.5025 t=0.5s g=—	E -2697.7631 t=0.9s g=—	I -92.0954 t=24.6s g=—	-	—	~ -384.5025 (nlopt, TL)
ex8_3_1	I -0.8041 t=4.0s g=—	I -0.7950 t=3.1s g=—	OK 0.0000 t=0.4s g=0.00e+00	E -0.1140 t=3.2s g=—	F 0.0000 t=17.5s g=—	nlopt	0.0000	OK 0.0000 (nlopt)
ex8_3_10	OK -1.2824 t=2.6s g=0.00e+00	F -0.0587 t=5.6s g=—	OK -0.0389 t=2.0s g=9.70e-01	OK -1.0036 t=2.0s g=2.17e-01	E -0.0360 t=16.9s g=—	ipopt	-1.2824	OK -1.2824 (ipopt)
ex8_3_12	OK -0.9938 t=0.8s g=0.00e+00	OK -0.9938 t=0.8s g=4.44e-16	OK 0.0000 t=0.5s g=9.94e-01	E -3.4576 t=2.7s g=—	F 0.0000 t=17.6s g=—	ipopt	-0.9938	OK -0.9938 (ipopt)
ex8_3_13	OK -42.6535 t=0.7s g=0.00e+00	OK -36.0847 t=0.6s g=1.54e-01	OK 0.0000 t=0.5s g=1.00e+00	E 2.500e+04 t=2.4s g=—	I 0.0000 t=19.3s g=—	ipopt	-42.6535	OK -42.6535 (ipopt)
ex8_3_14	F -1.4375 t=0.6s g=—	I — t=3.6s g=—	I -100.0000 t=4.0s g=—	OK -1.3976 t=2.1s g=0.00e+00	F -0.7925 t=18.7s g=—	optim	-1.3976	OK -1.3976 (optim)
ex8_3_2	OK -0.4123 t=0.6s g=3.83e-08	OK -0.4123 t=0.5s g=0.00e+00	OK 0.0000 t=0.5s g=4.12e-01	E -0.0083 t=2.9s g=—	X — t=25.2s g=—	madnlp	-0.4123	OK -0.4123 (madnlp)
ex8_3_3	OK -0.3330 t=0.5s g=8.36e-02	OK -0.4166 t=0.6s g=0.00e+00	OK 0.0000 t=0.5s g=4.17e-01	E -1.784e-05 t=2.5s g=—	OK -0.3330 t=10.7s g=8.36e-02	madnlp	-0.4166	OK -0.4166 (madnlp)
ex8_3_4	OK -3.5800 t=0.7s g=0.00e+00	OK -3.0607 t=0.5s g=1.45e-01	OK 0.0000 t=0.4s g=1.00e+00	E -2.3905 t=2.4s g=—	X — t=37.5s g=—	ipopt	-3.5800	OK -3.5800 (ipopt)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
ex8_3_5	OK -0.0627 t=0.6s g=6.47e-03	OK -0.0691 t=0.7s g=0.00e+00	OK 0.0000 t=0.4s g=6.91e-02	E -5.0930 t=2.4s g=—	F 0.0000 t=9.5s g=—	madnlp	-0.0691	OK -0.0691 (madnlp)
ex8_3_6	OK -0.5000 t=0.7s g=3.64e-09	OK -0.5000 t=0.5s g=0.00e+00	OK -0.0000 t=0.4s g=5.00e-01	E 1.0237 t=3.3s g=—	F -0.0000 t=17.9s g=—	madnlp	-0.5000	OK -0.5000 (madnlp)
ex8_3_7	E -0.8910 t=0.9s g=—	E -1.2126 t=1.3s g=—	E -3.144e+13 t=0.6s g=—	OK -1.2245 t=0.6s g=0.00e+00	OK -1.1236 t=16.0s g=8.24e-02	optim	-1.2245	OK -1.2245 (optim)
ex8_3_8	F -0.0124 t=0.6s g=—	OK -3.2561 t=0.7s g=3.72e-01	OK 0.0000 t=0.5s g=1.00e+00	OK -5.1811 t=0.6s g=0.00e+00	F 0.0000 t=12.8s g=—	optim	-5.1811	OK -5.1811 (optim)
ex8_3_9	OK -0.7416 t=0.4s g=9.23e-01	OK -0.7416 t=0.3s g=9.23e-01	OK 0.0000 t=0.3s g=1.00e+00	OK -9.6273 t=0.4s g=0.00e+00	F 0.0000 t=9.9s g=—	optim	-9.6273	OK -9.6273 (optim)
ex8_4_1	OK 0.6186 t=0.1s g=1.23e-13	OK 0.6186 t=0.1s g=0.00e+00	OK 0.6186 t=0.1s g=4.04e-10	OK 0.8895 t=0.1s g=2.71e-01	X — t=25.2s g=—	madnlp	0.6186	OK 0.6186 (madnlp)
ex8_4_2	OK 0.4852 t=0.2s g=0.00e+00	OK 0.4852 t=0.1s g=6.39e-14	OK 0.4852 t=0.1s g=1.01e-08	OK 0.8903 t=0.1s g=4.05e-01	X — t=32.4s g=—	ipopt	0.4852	OK 0.4852 (ipopt)
ex8_4_3	OK 0.0046 t=0.2s g=0.00e+00	OK 0.0046 t=0.2s g=3.91e-13	OK 0.0046 t=0.2s g=3.02e-10	E 1.8005 t=0.6s g=—	OK 0.0046 t=22.2s g=1.87e-11	ipopt	0.0046	OK 0.0046 (ipopt)
ex8_4_4	OK 0.2125 t=0.1s g=9.71e-14	OK 0.2125 t=0.1s g=0.00e+00	OK 0.2125 t=0.1s g=4.71e-09	E 0.2654 t=0.2s g=—	OK 0.2125 t=22.3s g=1.21e-11	madnlp	0.2125	OK 0.2125 (madnlp)
ex8_4_5	OK 3.075e-04 t=0.1s g=3.40e-11	OK 3.075e-04 t=0.1s g=3.40e-11	OK 0.0020 t=0.1s g=1.67e-03	OK 0.0015 t=0.1s g=1.22e-03	OK 3.075e-04 t=20.9s g=0.00e+00	uno	3.075e-04	OK 3.075e-04 (uno)
ex8_4_6	OK 0.0011 t=0.3s g=5.05e-13	OK 0.0011 t=0.2s g=0.00e+00	I 326.5159 t=0.2s g=—	OK 20.6513 t=0.2s g=2.07e+01	E 1.709e+21 t=16.0s g=—	madnlp	0.0011	OK 0.0011 (madnlp)
ex8_4_7	OK 29.0473 t=0.3s g=2.90e+01	OK 29.0473 t=0.3s g=2.90e+01	OK 29.0473 t=0.3s g=2.90e+01	OK 9.436e-04 t=0.3s g=0.00e+00	X — t=38.3s g=—	optim	9.436e-04	OK 9.436e-04 (optim)
ex8_4_8	OK 3.3218 t=0.4s g=6.22e-09	OK 3.3218 t=0.8s g=3.94e-09	OK 146.3400 t=0.8s g=4.31e+01	OK 18.3960 t=0.9s g=4.54e+00	OK 3.3218 t=16.3s g=0.00e+00	uno	3.3218	OK 3.3218 (uno)
ex8_5_1	OK -4.072e-07 t=0.1s g=7.79e-09	OK -4.072e-07 t=0.4s g=7.79e-09	OK -4.150e-07 t=2.6s g=0.00e+00	OK 2.0946 t=0.6s g=2.09e+00	E 7.1816 t=16.5s g=—	nlopt	-4.150e-07	OK -4.150e-07 (nlopt)
ex8_5_2	OK -1.150e-08 t=0.1s g=0.00e+00	OK -1.150e-08 t=0.1s g=2.03e-12	I 2.550e-06 t=0.1s g=—	OK 1.3086 t=0.0s g=1.31e+00	E 6.9269 t=16.8s g=—	ipopt	-1.150e-08	OK -1.150e-08 (ipopt)
ex8_5_3	OK -0.0041 t=0.0s g=0.00e+00	OK -0.0041 t=0.2s g=4.00e-16	I 0.0156 t=0.2s g=—	OK 20.3973 t=0.1s g=2.04e+01	F 0.0156 t=16.5s g=—	ipopt	-0.0041	OK -0.0041 (ipopt)
ex8_5_4	OK 1.005e-06 t=0.1s g=4.26e-04	OK 1.005e-06 t=0.0s g=4.26e-04	I 7.6733 t=0.0s g=—	OK 8.2418 t=0.0s g=8.24e+00	OK -4.251e-04 t=16.4s g=0.00e+00	uno	-4.251e-04	OK -4.251e-04 (uno)
ex8_5_5	OK -0.0108 t=0.1s g=0.00e+00	OK -0.0108 t=0.2s g=6.99e-16	OK 6.567e-07 t=0.2s g=1.08e-02	OK 6.736e-07 t=0.2s g=1.08e-02	OK 6.567e-07 t=16.3s g=1.08e-02	ipopt	-0.0108	OK -0.0108 (ipopt)
ex8_5_6	OK -0.0012 t=0.1s g=0.00e+00	OK -0.0012 t=0.0s g=3.40e-13	I 10.7672 t=0.0s g=—	OK 11.6261 t=0.0s g=1.16e+01	E 0.2404 t=16.8s g=—	ipopt	-0.0012	OK -0.0012 (ipopt)
ex8_6_1	OK -26.4905 t=0.5s g=3.87e-02	OK -27.5559 t=8.7s g=0.00e+00	OK -27.1149 t=0.6s g=1.60e-02	OK -27.4687 t=2.0s g=3.16e-03	OK -26.4644 t=17.0s g=3.96e-02	madnlp	-27.5559	OK -27.5559 (madnlp)
ex8_6_2	E 0.0000 t=0.2s g=—	E — t=0.2s g=—	I 1.635e+04 t=0.2s g=—	E — t=0.2s g=—	E 1.635e+04 t=14.0s g=—	-	—	~ 1.635e+04 (nlopt, TL)
ex9_1_1	E -12.9613 t=0.1s g=—	OK -13.0000 t=0.1s g=1.32e-01	OK 0.0000 t=0.1s g=1.00e+00	OK -14.9765 t=0.1s g=0.00e+00	OK -13.0000 t=9.4s g=1.32e-01	optim	-14.9765	OK -14.9765 (optim)
ex9_1_10	OK -3.2500 t=0.2s g=1.70e-06	OK -3.2500 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=1.00e+00	E — t=0.6s g=—	OK -3.2500 t=9.4s g=3.06e-09	madnlp	-3.2500	OK -3.2500 (madnlp)
ex9_1_2	OK -16.0000 t=0.1s g=0.00e+00	OK -16.0000 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=1.00e+00	OK -13.9567 t=0.1s g=1.28e-01	OK -16.0000 t=9.6s g=0.00e+00	madnlp	-16.0000	OK -16.0000 (madnlp)
ex9_1_3	OK -58.0000 t=0.2s g=1.11e-14	OK -58.0000 t=0.2s g=0.00e+00	OK -58.0000 t=0.1s g=2.00e-08	OK -58.0000 t=0.1s g=2.00e-08	OK -58.0000 t=9.4s g=2.00e-08	madnlp	-58.0000	OK -58.0000 (madnlp)
ex9_1_4	OK -37.0000 t=0.1s g=1.67e-11	OK -37.0000 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=1.00e+00	OK -5.3333 t=0.1s g=8.56e-01	F -38.7500 t=10.7s g=—	madnlp	-37.0000	OK -37.0000 (madnlp)
ex9_1_5	OK 4.0000 t=0.1s g=5.00e+00	OK 4.0000 t=0.1s g=5.00e+00	OK 0.0000 t=0.1s g=1.00e+00	OK 4.0000 t=0.1s g=5.00e+00	OK -1.0000 t=9.0s g=0.00e+00	uno	-1.0000	OK -1.0000 (uno)
ex9_1_6	OK -52.0000 t=0.1s g=0.00e+00	OK -52.0000 t=0.1s g=1.37e-16	OK -52.0000 t=0.1s g=1.92e-10	OK -52.0000 t=0.1s g=1.92e-10	OK -52.0000 t=9.2s g=1.92e-10	ipopt	-52.0000	OK -52.0000 (ipopt)
ex9_1_7	OK -50.0000 t=0.1s g=4.26e-16	OK -50.0000 t=0.1s g=0.00e+00	OK -50.0000 t=0.1s g=2.17e-08	OK -50.0000 t=0.1s g=2.17e-08	OK -50.0000 t=9.2s g=2.17e-08	madnlp	-50.0000	OK -50.0000 (madnlp)
ex9_1_8	OK -3.2500 t=0.1s g=1.70e-06	OK -3.2500 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=1.00e+00	E — t=0.5s g=—	OK -3.2500 t=9.4s g=3.06e-09	madnlp	-3.2500	OK -3.2500 (madnlp)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
ex9_1_9	OK 2.0000 t=0.1s g=0.00e+00	OK 2.0000 t=0.1s g=1.35e-13	OK 2.0000 t=0.1s g=2.31e-09	OK 2.0000 t=0.1s g=2.31e-09	OK 2.0000 t=9.4s g=2.31e-09	ipopt	2.0000	OK 2.0000 (ipopt)
ex9_2_1	OK 25.0000 t=0.1s g=4.71e-01	OK 25.0000 t=0.1s g=4.71e-01	OK 26.0000 t=0.1s g=5.29e-01	OK 20.3653 t=0.1s g=1.98e-01	OK 17.0000 t=9.8s g=0.00e+00	uno	17.0000	OK 17.0000 (uno)
ex9_2_2	OK 100.0000 t=0.1s g=9.70e-01	OK 100.0000 t=0.1s g=9.70e-01	OK 100.0000 t=0.1s g=9.70e-01	OK 50.7600 t=0.1s g=0.00e+00	E 100.0000 t=9.7s g=—	optim	50.7600	OK 50.7600 (optim)
ex9_2_3	OK 5.0000 t=0.1s g=1.08e+00	OK 5.0000 t=0.1s g=1.08e+00	OK -60.0000 t=0.1s g=0.00e+00	OK 11.2147 t=0.2s g=1.19e+00	OK 5.0000 t=9.4s g=1.08e+00	nlopt	-60.0000	OK -60.0000 (nlopt)
ex9_2_4	OK 0.5000 t=0.1s g=2.33e-10	OK 0.5000 t=0.0s g=2.33e-10	OK 4.0000 t=0.0s g=3.50e+00	OK 0.5000 t=0.1s g=2.33e-10	OK 0.5000 t=9.8s g=0.00e+00	uno	0.5000	OK 0.5000 (uno)
ex9_2_5	OK 9.8000 t=0.1s g=8.89e-02	OK 9.8000 t=0.1s g=8.89e-02	OK 13.0000 t=0.1s g=4.44e-01	OK 9.0000 t=0.1s g=0.00e+00	OK 9.0000 t=9.9s g=0.00e+00	optim	9.0000	OK 9.0000 (optim)
ex9_2_6	OK -1.0000 t=0.1s g=3.19e-09	OK -1.0000 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=1.00e+00	OK -0.4128 t=0.1s g=5.87e-01	OK -1.0000 t=9.8s g=8.18e-09	madnlp	-1.0000	OK -1.0000 (madnlp)
ex9_2_7	OK 25.0000 t=0.1s g=4.71e-01	OK 25.0000 t=0.1s g=4.71e-01	OK 26.0000 t=0.1s g=5.29e-01	OK 20.3653 t=0.1s g=1.98e-01	OK 17.0000 t=10.5s g=0.00e+00	uno	17.0000	OK 17.0000 (uno)
ex9_2_8	OK 1.5000 t=0.1s g=5.00e-01	OK 1.5000 t=0.1s g=5.00e-01	OK 1.0000 t=0.0s g=0.00e+00	OK 1.5000 t=0.0s g=5.00e-01	OK 1.5000 t=9.6s g=5.00e-01	nlopt	1.0000	OK 1.0000 (nlopt)
gtm	OK 543.5651 t=0.3s g=2.11e-01	OK 543.5651 t=0.2s g=2.11e-01	OK 543.5651 t=0.2s g=2.11e-01	OK 448.8228 t=0.3s g=0.00e+00	OK 543.5651 t=16.0s g=2.11e-01	optim	448.8228	OK 448.8228 (optim)
harker	OK -986.5135 t=0.1s g=0.00e+00	OK -970.9470 t=0.1s g=1.58e-02	OK -984.9107 t=0.1s g=1.62e-03	OK -519.9692 t=0.1s g=4.73e-01	OK -986.5135 t=15.3s g=3.27e-10	ipopt	-986.5135	OK -986.5135 (ipopt)
haverly	OK -400.0000 t=0.1s g=0.00e+00	OK -400.0000 t=0.1s g=2.87e-12	OK 0.0000 t=0.1s g=1.00e+00	E — t=0.5s g=—	OK 0.0000 t=9.5s g=1.00e+00	ipopt	-400.0000	OK -400.0000 (ipopt)
hhfair	OK -87.1590 t=0.2s g=4.38e-11	OK -87.1590 t=0.1s g=4.38e-11	OK -87.1590 t=0.2s g=0.00e+00	E -1.1002 t=0.3s g=—	I -91.6222 t=19.0s g=—	nlopt	-87.1590	OK -87.1590 (nlopt)
himmel11	OK -3.067e+04 t=0.0s g=4.82e-02	OK -3.067e+04 t=0.1s g=4.82e-02	OK -3.067e+04 t=0.0s g=4.82e-02	OK -3.222e+04 t=0.1s g=0.00e+00	I -3.067e+04 t=9.8s g=—	optim	-3.222e+04	OK -3.222e+04 (optim)
himmel16	OK 0.0000 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=0.00e+00	OK 0.0000 t=0.1s g=0.00e+00	OK 0.0000 t=8.9s g=0.00e+00	nlopt	0.0000	OK 0.0000 (nlopt)
house	OK -4500.0000 t=0.1s g=0.00e+00	OK -4500.0000 t=0.1s g=0.00e+00	OK -4500.0000 t=0.1s g=1.01e-08	E — t=0.4s g=—	OK -4500.0000 t=9.2s g=1.01e-08	ipopt	-4500.0000	OK -4500.0000 (ipopt)
hydro	OK 4.367e+06 t=0.2s g=1.60e-09	OK 4.367e+06 t=0.2s g=0.00e+00	OK 4.367e+06 t=0.2s g=1.11e-06	OK 4.391e+06 t=0.2s g=5.45e-03	OK 4.367e+06 t=9.2s g=1.81e-08	madnlp	4.367e+06	OK 4.367e+06 (madnlp)
immun	OK -9.537e-07 t=0.1s g=0.00e+00	OK -9.537e-07 t=0.1s g=0.00e+00	OK -4.768e-07 t=0.1s g=1.77e-07	E -3.815e-06 t=0.2s g=—	OK 0.0000 t=8.5s g=9.54e-07	ipopt	-9.537e-07	OK -9.537e-07 (ipopt)
launch	OK 2257.7973 t=0.2s g=9.48e-11	OK 2257.7973 t=0.2s g=0.00e+00	OK 2257.7976 t=0.2s g=1.01e-07	E 530.1394 t=0.4s g=—	E 2257.7976 t=17.5s g=—	madnlp	2257.7973	OK 2257.7973 (madnlp)
least	OK 1.409e+04 t=0.1s g=0.00e+00	OK 1.409e+04 t=0.0s g=1.42e-15	OK 1.100e+05 t=0.0s g=6.81e+00	OK 2.467e+04 t=0.1s g=7.52e-01	E 2.468e+04 t=13.9s g=—	ipopt	1.409e+04	OK 1.409e+04 (ipopt)
linear	E 541.0155 t=0.2s g=—	S 89.4363 t=1.4s g=—	OK 89.9006 t=0.2s g=0.00e+00	OK 738.0881 t=0.3s g=7.21e+00	E 89.2564 t=14.3s g=—	nlopt	89.9006	OK 89.9006 (nlopt)
Ints100	OK 0.5546 t=13.3s g=5.55e-01	OK 0.5546 t=2.2s g=5.55e-01	OK 0.0000 t=7.2s g=0.00e+00	OK 0.4521 t=2.6s g=4.52e-01	OK 0.5546 t=26.3s g=5.55e-01	nlopt	0.0000	OK 0.0000 (nlopt)
Ints50	OK 0.5547 t=1.1s g=5.55e-01	OK 0.5547 t=1.2s g=5.55e-01	OK 0.0000 t=2.1s g=0.00e+00	OK 0.4521 t=1.4s g=4.52e-01	OK 0.5547 t=19.4s g=5.55e-01	nlopt	0.0000	OK 0.0000 (nlopt)
meanvar	OK 5.2434 t=0.1s g=9.56e-12	OK 5.2434 t=0.1s g=0.00e+00	OK 5.2434 t=0.0s g=1.13e-08	OK 22.9333 t=0.1s g=3.37e+00	OK 5.2434 t=8.7s g=1.13e-08	madnlp	5.2434	OK 5.2434 (madnlp)
mhw4d	OK 0.0293 t=0.0s g=6.93e-14	OK 0.0293 t=0.0s g=0.00e+00	OK 1.0000 t=0.0s g=9.71e-01	OK 0.4176 t=0.0s g=3.88e-01	OK 0.0293 t=16.7s g=4.10e-14	madnlp	0.0293	OK 0.0293 (madnlp)
minlphi	OK 582.2361 t=0.3s g=1.11e-09	OK 582.2361 t=1.4s g=0.00e+00	OK 582.2361 t=0.3s g=3.57e-09	OK 582.2361 t=4.6s g=3.57e-09	OK 582.2361 t=16.5s g=3.57e-09	madnlp	582.2361	OK 582.2361 (madnlp)
nemhaus	OK 31.0000 t=0.0s g=0.00e+00	E — t=0.0s g=—	OK 31.0000 t=0.0s g=0.00e+00	OK 31.0000 t=0.0s g=0.00e+00	OK 31.0000 t=7.8s g=0.00e+00	optim	31.0000	OK 31.0000 (optim)
otpop	OK -7.390e-13 t=0.4s g=0.00e+00	OK -7.390e-13 t=0.4s g=0.00e+00	OK 4.083e-09 t=0.6s g=4.08e-09	OK 1573.2632 t=0.4s g=1.57e+03	OK 8.527e-13 t=17.2s g=1.59e-12	ipopt	-7.390e-13	OK -7.390e-13 (ipopt)
pindyck	OK -1612.1783 t=0.5s g=0.00e+00	OK -1612.1783 t=0.5s g=1.41e-16	OK -1612.1783 t=1.1s g=3.09e-08	OK 12.9056 t=0.5s g=1.01e+00	I 572.1115 t=18.0s g=—	ipopt	-1612.1783	OK -1612.1783 (ipopt)
pollut	OK -5.353e+06 t=0.2s g=9.19e-09	OK -5.353e+06 t=0.2s g=0.00e+00	OK -5.353e+06 t=0.3s g=4.08e-07	E — t=1.4s g=—	OK -5.353e+06 t=15.4s g=1.07e-08	madnlp	-5.353e+06	OK -5.353e+06 (madnlp)

Problema	ipopt	madnlp	nlopt	optim	uno	Melhor	f_best	Melhor situação
polygon100	OK -0.7197 t=18.9s g=0.00e+00	OK -0.6750 t=20.4s g=4.48e-02	OK -0.6750 t=27.6s g=4.48e-02	E — t=82.0s g=—	OK -0.6572 t=31.6s g=6.26e-02	ipopt	-0.7197	OK -0.7197 (ipopt)
polygon25	OK -0.7269 t=1.3s g=4.66e-08	OK -0.7269 t=1.4s g=0.00e+00	OK -0.0000 t=1.3s g=7.27e-01	E — t=11.5s g=—	OK -0.0000 t=19.0s g=7.27e-01	madnlp	-0.7269	OK -0.7269 (madnlp)
polygon50	OK -0.6750 t=5.0s g=4.81e-09	OK -0.6750 t=7.1s g=4.81e-09	OK -0.6750 t=5.5s g=0.00e+00	E — t=64.7s g=—	OK -0.6572 t=18.6s g=1.78e-02	nlopt	-0.6750	OK -0.6750 (nlopt)
process	OK -1161.3366 t=0.1s g=1.79e-09	OK -1161.3366 t=0.1s g=0.00e+00	OK -1161.3366 t=0.1s g=9.99e-09	OK -1161.3366 t=0.1s g=1.00e-08	OK -1161.3366 t=18.8s g=9.99e-09	madnlp	-1161.3366	OK -1161.3366 (madnlp)
prolog	OK 4.630e-07 t=0.2s g=3.23e-09	OK 4.597e-07 t=0.1s g=0.00e+00	OK 5.7486 t=0.1s g=5.75e+00	OK 369.1798 t=0.2s g=3.69e+02	OK 1.0622 t=9.5s g=1.06e+00	madnlp	4.597e-07	OK 4.597e-07 (madnlp)
qp1	OK 8.093e-04 t=0.6s g=2.75e-08	OK 8.094e-04 t=0.6s g=8.59e-08	OK 8.093e-04 t=0.7s g=1.64e-12	E 8.727e-04 t=2.8s g=—	OK 8.093e-04 t=10.9s g=0.00e+00	uno	8.093e-04	OK 8.093e-04 (uno)
qp2	OK 8.093e-04 t=0.6s g=2.75e-08	OK 8.094e-04 t=0.7s g=8.59e-08	OK 8.093e-04 t=0.7s g=1.64e-12	E 8.344e-04 t=3.1s g=—	OK 8.093e-04 t=10.1s g=0.00e+00	uno	8.093e-04	OK 8.093e-04 (uno)
qp3	OK 8.094e-04 t=0.9s g=8.71e-08	OK 8.094e-04 t=0.9s g=8.71e-08	OK 8.093e-04 t=1.1s g=5.90e-12	E — t=6.3s g=—	OK 8.093e-04 t=9.7s g=0.00e+00	uno	8.093e-04	OK 8.093e-04 (uno)
qp4	OK 8.094e-04 t=0.6s g=8.71e-08	OK 8.094e-04 t=0.6s g=8.71e-08	OK 8.093e-04 t=0.7s g=4.07e-10	OK 8.227e-04 t=0.7s g=1.33e-05	OK 8.093e-04 t=9.8s g=0.00e+00	uno	8.093e-04	OK 8.093e-04 (uno)
qp5	OK 0.4315 t=0.7s g=5.43e-12	OK 0.4315 t=0.8s g=0.00e+00	OK 0.4315 t=0.7s g=1.65e-06	OK 0.4315 t=1.2s g=1.65e-06	OK 0.4315 t=8.9s g=1.65e-06	madnlp	0.4315	OK 0.4315 (madnlp)
ramsey	OK -2.4875 t=0.2s g=0.00e+00	OK -2.4875 t=0.1s g=1.52e-12	OK -2.4875 t=0.1s g=7.78e-08	OK -2.4875 t=0.2s g=3.09e-08	OK -2.4875 t=16.3s g=3.09e-08	ipopt	-2.4875	OK -2.4875 (ipopt)
rbrock	OK 8.249e-24 t=0.0s g=6.66e-24	OK 1.585e-24 t=0.0s g=0.00e+00	OK 3.242e-22 t=0.0s g=3.23e-22	OK 2.217e-23 t=0.0s g=2.06e-23	OK 1.168e-15 t=14.3s g=1.17e-15	madnlp	1.585e-24	OK 1.585e-24 (madnlp)
robot50	E -8.035e+21 t=3.4s g=—	I -6.911e+26 t=12.9s g=—	I -1.600e+04 t=538.4s g=—	OK -3748.7500 t=2.6s g=0.00e+00	F 0.0000 t=20.9s g=—	optim	-3748.7500	OK -3748.7500 (optim)
rocket100	OK -1.0128 t=3.0s g=6.85e-01	OK -1.0128 t=2.9s g=6.85e-01	OK -1.0000 t=10.2s g=6.89e-01	OK -3.2183 t=7.6s g=0.00e+00	F -1.0000 t=19.6s g=—	optim	-3.2183	OK -3.2183 (optim)
rocket50	OK -1.0128 t=1.6s g=6.83e-01	OK -1.0128 t=1.4s g=6.83e-01	OK -1.0000 t=1.9s g=6.87e-01	OK -3.1946 t=2.0s g=0.00e+00	F -1.0000 t=17.7s g=—	optim	-3.1946	OK -3.1946 (optim)
sambal	OK 3.9682 t=0.1s g=4.15e-12	OK 3.9682 t=0.1s g=3.58e-15	OK 331.0368 t=0.1s g=8.24e+01	E 3.9685 t=0.2s g=—	OK 3.9682 t=8.3s g=0.00e+00	uno	3.9682	OK 3.9682 (uno)
sample	OK 726.6789 t=0.0s g=4.39e-12	OK 726.6789 t=0.0s g=0.00e+00	OK 726.6794 t=0.0s g=6.99e-07	OK 726.6794 t=0.0s g=6.76e-07	OK 726.6794 t=15.0s g=6.76e-07	madnlp	726.6789	OK 726.6789 (madnlp)
ship	OK 5.5409 t=0.1s g=4.81e-16	OK 5.5409 t=0.9s g=0.00e+00	I 44.0043 t=1.1s g=—	OK 49.9120 t=1.0s g=8.01e+00	E 6.0034 t=18.1s g=—	madnlp	5.5409	OK 5.5409 (madnlp)
srcpm	OK 2109.7824 t=0.2s g=2.48e-07	OK 2109.7823 t=0.2s g=2.36e-07	OK 2109.7818 t=0.2s g=1.08e-09	OK 5549.4435 t=0.2s g=1.63e+00	OK 2109.7818 t=14.7s g=0.00e+00	uno	2109.7818	OK 2109.7818 (uno)
turkey	OK -2.933e+04 t=2.9s g=4.05e-10	OK -2.933e+04 t=2.7s g=0.00e+00	I -2.824e+04 t=484.1s g=—	E -451.7507 t=12.9s g=—	OK -2.933e+04 t=11.0s g=1.04e-09	madnlp	-2.933e+04	OK -2.933e+04 (madnlp)
wall	OK 1.0000 t=0.0s g=0.00e+00	OK 1.0000 t=0.4s g=0.00e+00	OK 1.0000 t=0.4s g=0.00e+00	OK 1.0000 t=0.2s g=0.00e+00	OK 1.0000 t=15.1s g=2.89e-15	ipopt	1.0000	OK 1.0000 (ipopt)
water	OK 927.1234 t=0.2s g=9.27e+02	OK 923.7109 t=0.3s g=9.24e+02	OK 0.0000 t=0.4s g=0.00e+00	OK 1712.3989 t=0.3s g=1.71e+03	I 867.2024 t=19.7s g=—	nlopt	0.0000	OK 0.0000 (nlopt)
weapons	OK -1735.5696 t=0.2s g=7.00e-13	OK -1735.5696 t=0.2s g=0.00e+00	OK -1735.5666 t=0.4s g=1.73e-06	E -1738.1403 t=0.7s g=—	OK -1735.5696 t=14.7s g=5.14e-10	madnlp	-1735.5696	OK -1735.5696 (madnlp)

5. Análise gráfica

Comparativo de desempenho dos cinco solvers. No perfil de desempenho (Dolan–Moré), curvas mais à esquerda/acima indicam solvers mais eficientes e robustos. Optim* = proxy Ipopt.

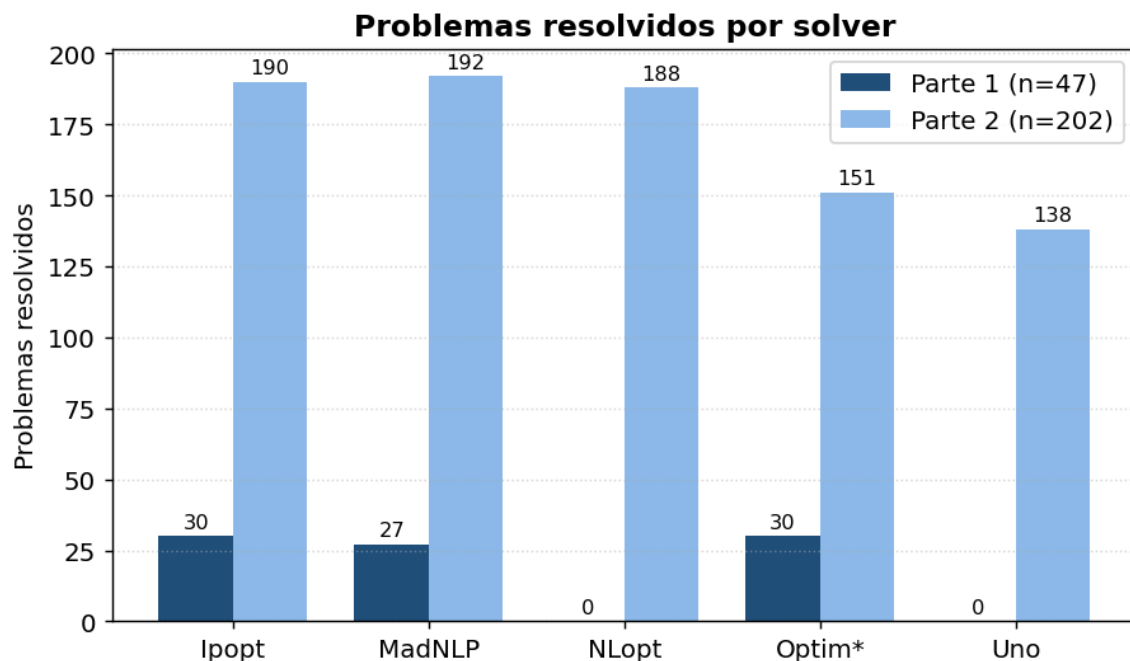


Figura 1 — Problemas resolvidos por solver (Parte 1 vs Parte 2).

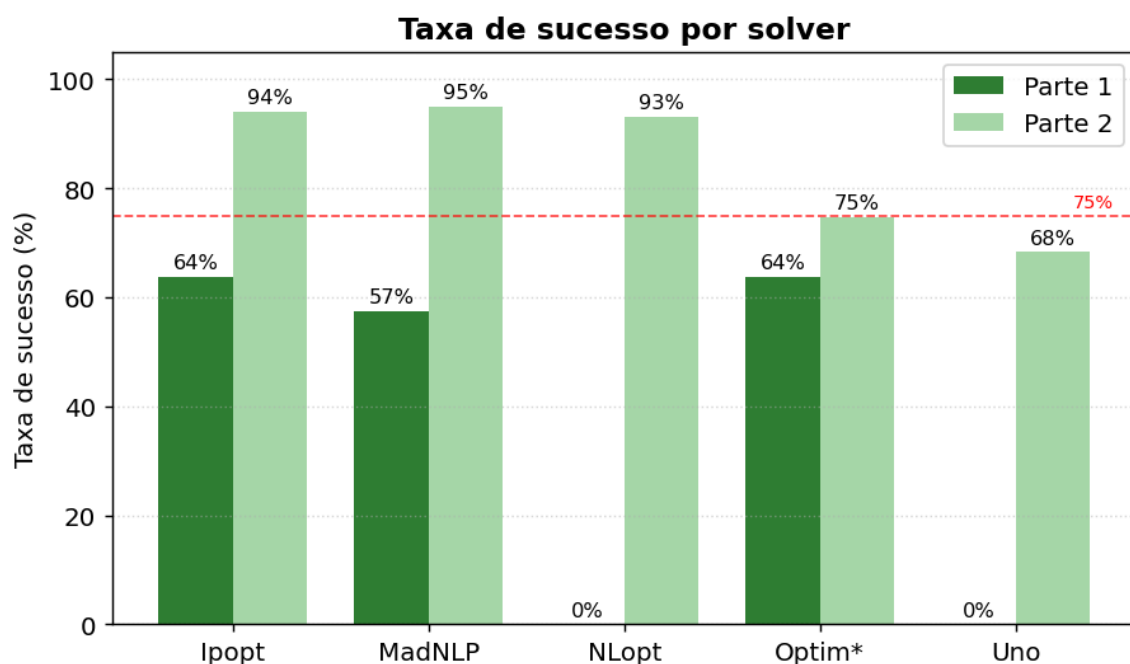


Figura 2 — Taxa de sucesso (%) por solver.

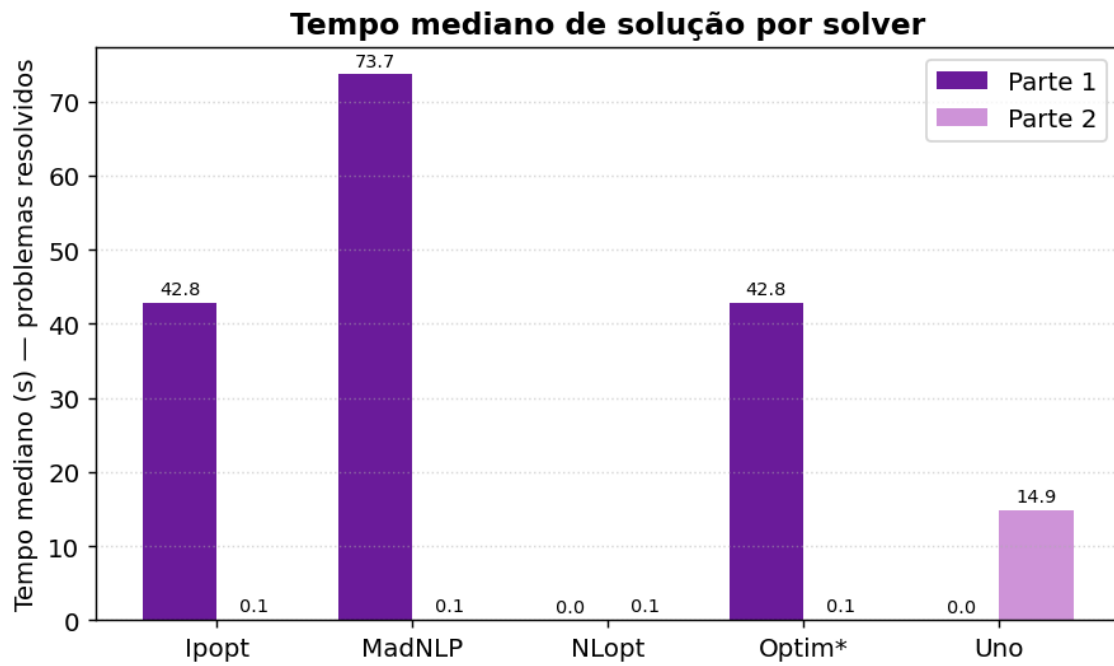


Figura 3 — Tempo mediano de solução por solver.

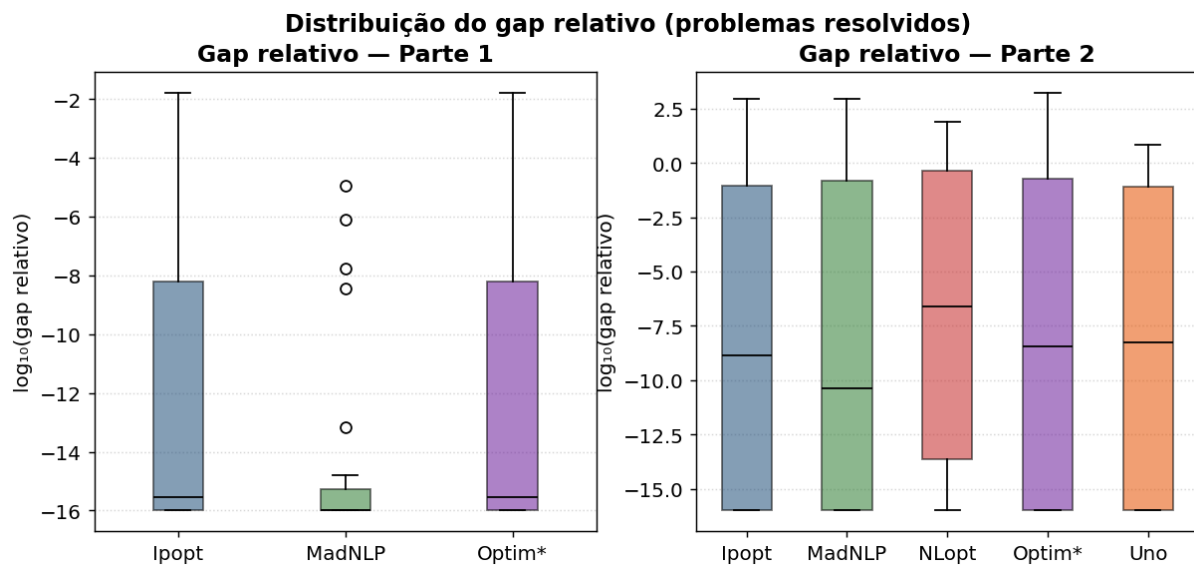


Figura 4 — Distribuição do gap relativo (log) dos resolvidos.

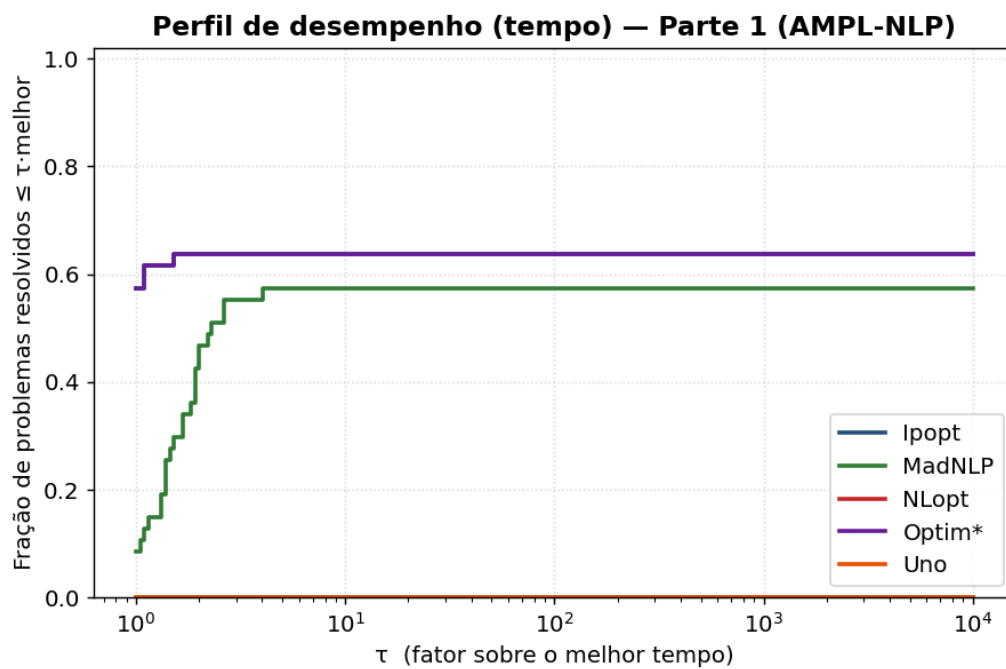


Figura 5 — Perfil de desempenho (Dolan–Moré) — Parte 1.

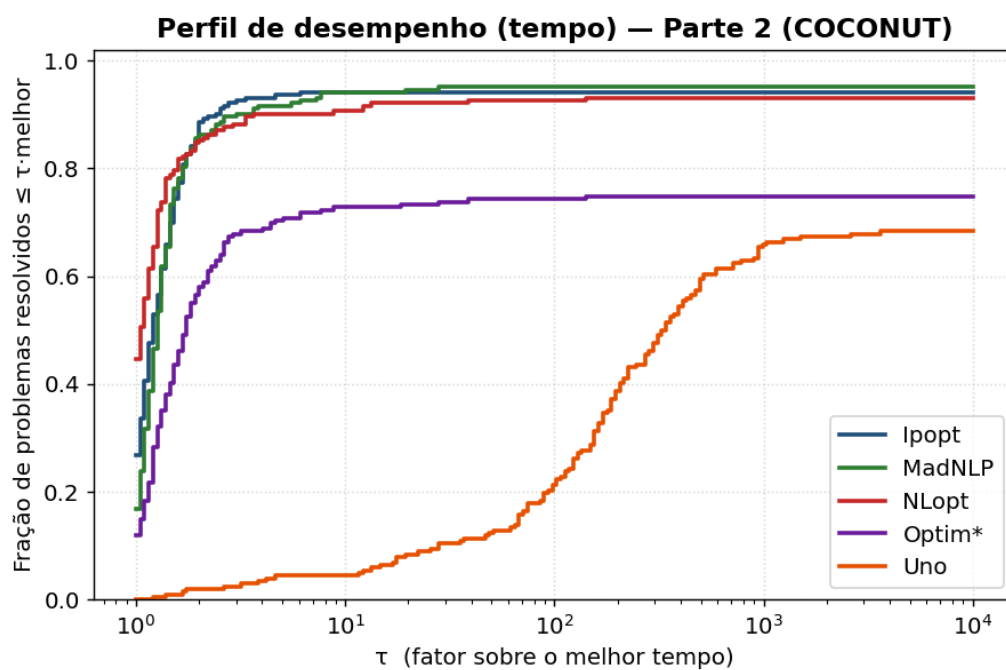


Figura 6 — Perfil de desempenho (Dolan–Moré) — Parte 2.

6. Conclusões

Parte 1 — taxa de sucesso: ipopt=30/47 (63.8%), madnlp=27/47 (57.4%), nlopt=0/47 (0.0%), optim=30/47 (63.8%), uno=0/47 (0.0%).

Parte 2 — taxa de sucesso: ipopt=190/202 (94.1%), madnlp=192/202 (95.0%), nlopt=188/202 (93.1%), optim=151/202 (74.8%), uno=138/202 (68.3%).

Cobertura na melhor situação — Parte 1: 31 resolvidos + 4 incumbentes = 35/47 (74%); 12 sem solução. Parte 2: 200 resolvidos + 2 incumbentes = 202/202 (100%).

Ipopt (e Optim*=proxy Ipopt) e MadNLP são os mais robustos: ~57–64% na Parte 1 e ~94–95% na Parte 2. NLOpt é forte na Parte 2 (93%) mas falha na Parte 1 (SLSQP denso estoura memória nos problemas grandes de controle ótimo). Uno resolve 68% da Parte 2 mas não converge no tempo nos grandes da Parte 1. A dificuldade da Parte 1 vem do porte/densidade das instâncias, favorecendo pontos interiores esparsos (Ipopt/MadNLP), como confirma o perfil de desempenho (Figuras 5–6).